

»» Chapter VII ««

FURTHER APPLICATIONS OF THE CAUCHY THEORY

PREFATORY REMARKS

The present chapter treats two consequences of the Cauchy theory fecund in their applications to certain calculations of classical analysis. One is the *residue theorem* relating under appropriate conditions line integrals of analytic functions with isolated singularities to the residues at the singularities. The proof of this theorem is simple. In situations where the residues are easy to compute, the identities afforded by the theorem may be used to calculate definite integrals (proper and improper) under favorable conditions, as we shall see by a study of examples. The application of the residue theorem for such purposes parallels (indeed it may be thought of as generalizing) “the method of the Cauchy integral theorem” used to calculate the Fresnel integrals and the other integrals treated in the exercises of Ch. V. As we shall see in the examples of §2, this chapter, successful application depends upon the combination of a “manageable” function, an astute choice of a path of integration, and care in the use of basic definitions. The problems will be found to offer good practice in the use of elementary complex function theory. An authoritative text on the residue theorem and its many applications is the useful monograph of Lindelöf [77].

The other consequence of the Cauchy theory to which we have referred is a corollary of the residue theorem and is known as “the method of Cauchy.” It gives an answer to the question: What do the Cauchy integral formulas yield

when the hypothesis of analyticity is relaxed to the extent of allowing a finite number of isolated singularities? The resulting compensated formula will lead to representation theorems for such meromorphic functions as the cotangent.

1. THE RESIDUE THEOREM

We shall content ourselves (following the precedent of Saks and Zygmund) with a residue theorem satisfying fairly simple hypotheses insofar as a fully proved theorem is concerned. For applications the theorem given is abundantly adequate. As in the exposition of Ch. V, we shall indicate how more general theorems may be obtained.

Let A be a holomorphically simply connected region in $\mathbb{C}$, let $a_1, \dots, a_n$ be n ($\in \mathbb{N}$) distinct points of A , let f be analytic on $A - \{a_1, \dots, a_n\}$, and let γ be a rectifiable closed curve lying in $A - \{a_1, \dots, a_n\}$. By the *residue theorem* we understand

Theorem 1.1: *Under the stated conditions*

$$\int_{\gamma} f dz = 2\pi i \sum_1^n O(a_k; \gamma) \operatorname{res}(a_k; f).$$

PROOF: Let P_k denote the principal part of f at a_k . We see that $g(z) = f(z) - \sum_1^n P_k(z)$ with domain $A - \{a_1, \dots, a_n\}$ has removable singularities at $a_1, \dots, a_n$. Noting that

$$\int_{\gamma} g dz = 0,$$

and using the fact that

$$\begin{aligned} \int_{\gamma} P_k dz &= 2\pi i O(a_k; \gamma) \operatorname{res}(a_k; P_k) \\ &= 2\pi i O(a_k; \gamma) \operatorname{res}(a_k; f), \quad k = 1, \dots, n, \end{aligned}$$

we conclude the residue theorem.

What may we expect when the hypothesis that A be holomorphically simply connected is dropped? When we refer to the situation considered at the end of §3, Ch. V (φ being univalent), but now allow f to have a finite number of isolated singularities, all in $\varphi(\operatorname{int} E)$, that is, if f is given as analytic on $O - X$ where X is a finite subset of $\varphi(\operatorname{int} E)$, we see that the use of the

auxiliary g is available and we conclude

$$\int_{\beta} f dz - \sum_1^n \int_{\beta_k} f dz = 2\pi i \varepsilon \sum_{a \in X} \text{res}(a; f),$$

where ε is the common value of $O(z; \beta) - \sum_1^n O(z; \beta_k)$, $z \in \varphi(\text{int } E)$. The only matter that is not settled is that, in fact, ε is either 1 or -1 .

EXERCISES

1.1. Develop a residue theorem in the geometric setting of Ex. 1.12, Ch. VI, isolated singularities being allowed.

1.2. Let $0 < \rho < r$. Let f be analytic on $\Delta(0; r)$. (a) Calculate

$$\oint_{C(0; \rho)} \frac{f(z)}{\prod_1^n (z - a_k)} dz$$

where $a_1, \dots, a_n \in \Delta(0; \rho)$, distinguishing between the case where the sequence $(a_k)_{1 \leq k \leq n}$ is univalent and the case where it is not. (b) Suppose that $a \in \Delta(0; \rho)$, that $f'(a) \neq 0$, that $f(z) \neq f(a)$ for $z \neq a$ satisfying $|z| \leq \rho$, and that $n \in \mathbb{N}$. Calculate

$$\oint_{C(0; \rho)} z^n \frac{f(z)}{f(z) - f(a)} dz.$$

1.3. Let g be meromorphic on A of Theorem 1.1 and suppose that $\{\partial_q(z) \neq 0\}$ is a finite subset of $A - \gamma$, γ being as in the theorem. Calculate with the aid of the theorem

$$\int_{\gamma} \frac{g'}{g} dz, \int_{\gamma} \frac{z g'(z)}{g(z)} dz.$$

Note that the former integral was treated in quite a different manner in §4, Ch. VI (Argument Principle).

1.4. Let X be a topological space. Let $a_0, \dots, a_n: X \rightarrow \mathbb{C}$, $n \in \mathbb{N}$, each being continuous at $q \in X$. Suppose that

$$Q(w, p) = w^{n+1} + \sum_0^n a_k(p) w^k.$$

Let $0 < r < +\infty$ and suppose that ω is the unique root of $Q(w, q) = 0$ in $\overline{\Delta(\omega; r)}$ and that $Q_w(\omega, q) \neq 0$. Show that, for p sufficiently near q ,

$$\frac{1}{2\pi i} \oint_{C(\omega; r)} \frac{w Q_w(w, p)}{Q(w, p)} dw$$

is the unique root of $Q(w, p) = 0$ in $\overline{\Delta(\omega; r)}$ and that subject to the restriction in question it defines a function continuous at q .

1.5. Let γ be a rectifiable closed curve in $\mathbb{C}$, let $a, b \in \mathbb{C} - \gamma$, let $m, n \in \mathbb{N}$. Calculate

$$\int_{\gamma} (z-a)^{-m}(z-b)^{-n} dz.$$

1.6. Summation formulas. The residue theorem permits us to evaluate limits of the form

$$\lim_{n \rightarrow \infty} \sum_{k=-n}^n \varepsilon(k) [P(k)]^{-1}$$

where P is a polynomial of degree ≥ 2 having no zeros in $\mathbb{Z}$ and either $\varepsilon(k) = 1$, $k \in \mathbb{Z}$, or $\varepsilon(k) = (-1)^k$. Develop the details of the following approach to the question: (a) With $0 < r < +\infty$ and f meromorphic on $\mathbb{C}$, let

$$M_1(r; f) = \sup \left\{ |f(z)| : z \in \mathbb{C} - \bigcup_{k \in \mathbb{Z}} \Delta(k; r) \right\},$$

and let

$$M_2(f) = \sup \left\{ |f(z)| : z \in \bigcup_{n \in \mathbb{N}} \text{fr} \left[-\left(n + \frac{1}{2}\right), n + \frac{1}{2}; -\left(n + \frac{1}{2}\right), n + \frac{1}{2} \right] \right\}.$$

Show that $M_1(r; f)$, $M_2(f) < +\infty$, when $f(z) = \cot \pi z$, $z \in \mathbb{C}$, or $f(z) = \csc \pi z$, $z \in \mathbb{C}$. (b) Apply the residue theorem to

$$A_n = \int_{\beta_n} \frac{\cot \pi z}{P(z)} dz$$

and

$$B_n = \int_{\beta_n} \frac{\csc \pi z}{P(z)} dz,$$

where β_n is the positively sensed boundary path of

$$\left[-\left(n + \frac{1}{2}\right), n + \frac{1}{2}; -\left(n + \frac{1}{2}\right), n + \frac{1}{2} \right],$$

n being large. (c) Show that $\lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} B_n = 0$ and conclude the desired summation formulas.

Apply to the case where $P(z) = z^2 + a$, $0 < a < +\infty$. Consider also the general question when P may have a zero in $\mathbb{Z}$ and interpret the consequences of $\lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} B_n = 0$. Apply to determine $\sum_0^{\infty} (k+1)^{-2(n+1)}$, $n \in \mathbb{N}$, obtaining the values of the sum explicitly for $n = 0, 1, 2$.

In the following theorem, forming the base of the method of Cauchy, the hypotheses and notations of Theorem 1.1, this section, are in effect.

Theorem 1.2: *Under the stated conditions*

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw = O(z; \gamma) f(z) - \sum_1^n O(a_k; \gamma) P_k(z), \quad z \in A - \{a_1, \dots, a_n\}.$$

PROOF: Let F be $f(w)/(w - z)$ with domain $A - \{z, a_1, \dots, a_n\}$. We have

$$\operatorname{res}(z; F) = f(z)$$

and, using $f(w) = g(w) + \sum_1^n P_k(w)$,

$$\operatorname{res}(a_k; F) = \operatorname{res}[a_k; P_k(w)(w - z)^{-1}] = -P_k(z), \quad k = 1, \dots, n.$$

The theorem follows on applying Theorem 1.1 to F .

It is immediate that on differentiating the identity of the theorem m times we obtain

$$\frac{m!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z)^{m+1}} dw = O(z; \gamma) f^{(m)}(z) - \sum_1^n O(a_k; \gamma) P_k^{(m)}(z).$$

EXERCISES

1.7. Furnish the details of the proof of the equality

$$\operatorname{res}[a_k; P_k(w)(w - z)^{-1}] = -P_k(z).$$

Hint: Apply Theorem 3.1, Ch. VI.

1.8. Apply Theorem 1.2 in the special case: $f(w) = \varphi(w)/Q(w)$ with φ analytic on A , $Q(w) = \prod_1^n (w - a_k)$, to conclude

$$\frac{1}{2\pi i} \int_{\gamma} \frac{Q(z)}{Q(w)} \frac{\varphi(w)}{w - z} dw = O(z; \gamma) \varphi(z) - \sum_1^n O(a_k; \gamma) \frac{\varphi(a_k)}{Q'(a_k)} \frac{Q(z)}{z - a_k},$$

a formula due to Hermite.

Show with the aid of this formula that if $A = \Delta(0; r)$, $(a_k): \mathbf{N} \rightarrow \Delta(0; r)$, (a_k) is univalent, and $\lim_{k \rightarrow \infty} a_k = 0$, then the sequence of polynomials $(P_n)_{n \in \mathbf{N}}$, where P_n is of degree at most n and satisfies the interpolation condition $P_n(a_k) = \varphi(a_k)$, $k = 0, \dots, n$, tends uniformly to φ in $\Delta(0; r)$. [To be utterly strict, " $(P_n|_A)_{n \in \mathbf{N}}$ " should be read for " $(P_n)_{n \in \mathbf{N}}$."]

Extend the first part of the exercise replacing Q by a polynomial of nonnegative degree whose zeros are all in A . Formulate and prove a corresponding interpolation theorem in this setting.

2. APPLICATION OF THE RESIDUE THEOREM TO THE CALCULATION OF INTEGRALS

We shall show by the study of several examples how the residue theorem may be employed to calculate certain integrals. It will be seen that in standard situations a suitable approach readily suggests itself. Skill in the use of the residue theorem for such calculations is a matter of practice and experience.

Example 1. We seek the value of

$$\int_{-\infty}^{+\infty} \frac{1}{x^2 + 1} dx. \quad (2.1)$$

This example was deliberately chosen because it is simple and because the result is well known from the elementary integral calculus. Indeed, the principal determination of the arctangent (=the inverse of $\tan [\] - \pi/2, \pi/2[$) is a primitive of the integrand. The value of the integral is seen to be π . Let us approach the question from the point of view of the residue theorem. We have to do with the rational function $f = 1/(z^2 + 1)$, $z \in \hat{\mathbb{C}}$. It has two poles, each simple, one at i , the other at $-i$. Given $1 < r < +\infty$, we introduce

$$I(r) = \int_{(-r)(r)}^{\rightarrow} f dz + \int_{\gamma_r} f dz,$$

where $\gamma_r(\theta) = re^{i\theta}$, $0 \leq \theta \leq \pi$, and conclude from the residue theorem that

$$I(r) = 2\pi i \operatorname{res}(i; f) = 2\pi i(1/2i) = \pi.$$

Note that the order of $-i$ with respect to the closed curve associated with the indicated line integrals is 0. The absolute value of the second integral does not exceed the length of γ_r times $\max_{\gamma_r} |f|$, that is, $\pi r/(r^2 - 1)$. We conclude that the value of (2.1) is π . It is to be noted that the Cauchy integral formula for a semicircle applied to $(z + i)^{-1}$ also yields the value of $I(r)$.

Example 2. Let $0 < \alpha < 1$. We seek the value of

$$\int_0^{+\infty} \frac{x^{\alpha-1}}{1+x} dx. \quad (2.2)$$

Here, of course,

$$x^{\alpha-1} = \exp[(\alpha-1) \log x], \quad 0 < x < +\infty.$$

It is clear that the integral is convergent. This fact will emerge also from the ensuing calculation. We introduce $\lambda_1 = \operatorname{Log}$, $\lambda_2 =$ the analytic logarithm of the identity map on $\mathbb{C} - \{t: t \geq 0\}$ which agrees with λ_1 on $\{\operatorname{Im} z > 0\}$, and $\lambda_3 =$ the analytic logarithm of the identity map on $\mathbb{C} - \{t: t \leq 0\}$ which agrees with λ_2 on $\{\operatorname{Im} z < 0\}$. We note that $\lambda_1(e^{i\theta}) = i\theta$, $0 \leq \theta < \pi$, $\lambda_2(e^{i\theta}) = i\theta$, $0 < \theta < 2\pi$, $\lambda_3(e^{i\theta}) = i\theta$, $\pi < \theta < 3\pi$. Hence $\lambda_3(z) = \lambda_1(z) + 2\pi i$, z in the domain of λ_3 . Let $0 < \rho < 1 < r < +\infty$. Consider

$$\sum_1^3 \int_{\gamma_k} \frac{\exp[(\alpha-1) \lambda_k(z)]}{1+z} dz, \quad (2.3)$$

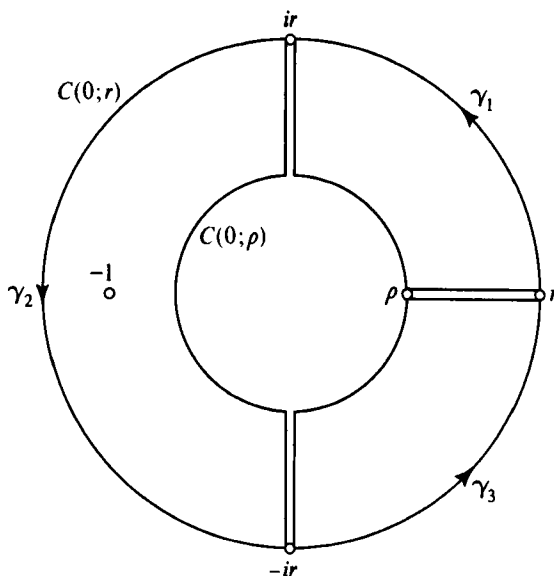


FIG. 7.1

where the γ_k are the positively sensed boundary paths indicated in the adjoining figure. The first and third integrals in the sum are zero, while by the residue theorem (or Cauchy integral formula) the value of the second is

$$2\pi i \exp[(\alpha - 1) \lambda_2(-1)] = -2\pi i e^{\alpha \pi i}.$$

On taking into account the relations among $\lambda_1, \lambda_2, \lambda_3$, we see that (2.3) is equal to the sum of

$$(1 - e^{2\alpha \pi i}) \int_{\rho}^r \frac{x^{\alpha-1}}{1+x} dx$$

and contributions, one from arcs on $C(0; \rho)$, the other from arcs on $C(0; r)$. The absolute value of the former contribution does not exceed

$$[\rho^{\alpha-1}(1-\rho)^{-1}](2\pi\rho),$$

and the absolute value of the latter contribution does not exceed

$$[r^{\alpha-1}(r-1)^{-1}](2\pi r).$$

Taking limits we conclude that the value of (2.2) is

$$\frac{2\pi i e^{\alpha \pi i}}{e^{2\alpha \pi i} - 1} = \frac{\pi}{\sin(\alpha \pi)}.$$

This example illustrates the care that is called for in the use of analytic logarithms. A second approach will be given in Ex. 2.5 below.

Example 3. Let $a \in \mathbb{C}$, $|a| \neq 1$. We seek the value of

$$\int_0^{2\pi} |e^{i\theta} - a|^{-2} d\theta. \quad (2.4)$$

We note that the integral may be represented as the line integral

$$\oint_{C(0;1)} \frac{1}{(z-a)(\bar{z}-\bar{a})} \cdot \frac{1}{iz} dz = \oint_{C(0;1)} \frac{1}{i(z-a)(1-\bar{a}z)} dz.$$

Using the residue theorem (or Cauchy integral formula) we see that the value of (2.4) is $2\pi/(1-|a|^2)$ when $|a| < 1$ and is $2\pi/(|a|^2-1)$ when $|a| > 1$.

An independent determination of the value of (2.4) when $|a| < 1$ is given by the Poisson integral for the real part of the constant one.

Example 4. Let $\alpha \in \mathbb{R}$, let $0 < c$, $t < +\infty$. We seek the value of $\lim_{+\infty} I$, where

$$I(t) = \int_{(c-it)(c+it)}^{\rightarrow} \frac{e^{az}}{z} dz. \quad (2.5)$$

When $\alpha \leq 0$, the Cauchy integral theorem for a semicircle serves. Indeed,

$$I(t) = \int_{-\pi/2}^{\pi/2} \frac{\exp[\alpha(c + te^{i\theta})]}{c + te^{i\theta}} ite^{i\theta} d\theta.$$

If $\alpha = 0$, $\lim_{+\infty} I = \pi i$. This result could also have been obtained by introducing an analytic logarithm of the identity map on $\{\operatorname{Re} z > 0\}$, which is a primitive of $1/z$. If $\alpha < 0$, we see from the estimate

$$\left| \int_{-\pi/2}^{\pi/2} \frac{\exp[\alpha(c + te^{i\theta})]}{c + te^{i\theta}} ite^{i\theta} d\theta \right| \leq \int_{-\pi/2}^{\pi/2} e^{\alpha t \cos \theta} d\theta$$

that $\lim_{+\infty} I = 0$.†

Suppose now that $\alpha > 0$. We take $t > c$ and note that by the residue theorem (or the Cauchy integral formula for a semicircle)

$$I(t) + \int_{\pi/2}^{3\pi/2} \frac{\exp[\alpha(c + te^{i\theta})]}{c + te^{i\theta}} ite^{i\theta} d\theta = 2\pi i \operatorname{res}\left(0; \frac{e^{az}}{z}\right) = 2\pi i.$$

Noting the estimate

$$\left| \int_{\pi/2}^{3\pi/2} \frac{\exp[\alpha(c + te^{i\theta})]}{c + te^{i\theta}} ite^{i\theta} d\theta \right| \leq \frac{t}{t-c} e^{\alpha c} \int_{\pi/2}^{3\pi/2} e^{\alpha t \cos \theta} d\theta$$

we conclude that $\lim_{+\infty} I = 2\pi i$.

† Cf. Ex. 3.24, Ch. V.

Integrals of the present type occur in the theory of the Laplace transform.

It is to be remarked that we could have carried out all the integrations of these examples with the aid of a Cauchy integral formula as well as with the aid of the residue theorem. This happened because we wished to keep our illustrative examples simple. In the exercises that follow the occurrence of more than one pole will not be unusual.

EXERCISES

In this set the indicated integrals and limits are to be evaluated.

2.1. (a) $\int_{-\infty}^{+\infty} \frac{1}{1+x^4} dx;$

(b) $\int_0^{+\infty} \frac{1}{1+x^3} dx;$

(c) $\int_{-\infty}^{+\infty} \frac{x^m}{1+x^{2n}} dx, m, n \in \mathbf{N}, 2n-m \geq 2;$

(d) $\int_{-\infty}^{+\infty} \frac{1}{(1+x^2)^{n+1}} dx, n \in \mathbf{N};$

(e) $\lim_{+\infty} I$, where $I(t) = \int_{-\infty}^t f(x) dx$, $f: \{\operatorname{Im} z \geq 0\} \rightarrow \hat{\mathbf{C}}$ being continuous on its domain, meromorphic in $\{\operatorname{Im} z > 0\}$, and satisfying $\lim_{\infty} z f(z) = 0$, $f(\mathbf{R}) \subset \mathbf{C}$;

(f) $\lim_{+\infty} I$, where $I(t) = \int_{-\infty}^t e^{i\alpha x} f(x) dx$, $0 < \alpha < +\infty$, f satisfying the above conditions save that the limit requirement is replaced by $\lim_{\infty} f = 0$;

(g) $\int_0^{+\infty} \frac{\sin \alpha x}{x(x^2+1)^2} dx, \alpha \in \mathbf{R}.$

2.2. * (a) $\int_0^{+\infty} \frac{\log x}{(1+x)^3} dx;$

(b) $\int_0^{+\infty} x^{\alpha}(1+x)^{-4} dx, -1 < \alpha < 3.$

2.3. (a) $\int_0^{2\pi} |P(e^{i\theta})|^{-2} d\theta$, P a polynomial having no zeros on $C(0; 1)$;

(b) $\int_0^{2\pi} [T(\theta)]^{-1} d\theta$ where $T(\theta) = \sum_0^n (A_k \cos k\theta + B_k \sin k\theta)$, $A_k, B_k \in \mathbf{R}$, $T(\theta) \neq 0$ for $\theta \in \mathbf{R}$;

(c) $\int_0^{2\pi} \frac{P(\cos \theta, \sin \theta)}{Q(\cos \theta, \sin \theta)} d\theta$ where $P(x, y)$ and $Q(x, y)$ are polynomials with real coefficients and $Q(\cos \theta, \sin \theta) \neq 0$ for $\theta \in \mathbf{R}$;

(d) $\int_{-\infty}^{+\infty} \frac{1}{|P(x)|^2} dx$, where P is a polynomial of positive degree with all its zeros simple and having positive imaginary part.

2.4. $\lim_{+\infty} I, I(t) = \int_{c-it}^{c+it} \frac{e^{\alpha z}}{z^{n+1}} dz$, where $\alpha \in \mathbf{R}, n \in \mathbf{N}, 0 < c < +\infty$.

2.5. $\int_{-\infty}^{+\infty} \frac{e^{\alpha t}}{1+e^t} dt, 0 < \alpha < 1$. [Consider the positively sensed boundary path of $[-r, r; 0, 2\pi], 0 < r < +\infty$.] This yields the result of Example 2 of the text as we see on making the substitution $x = e^t$.

2.6. $\int_0^\pi \log \sin \theta d\theta$ by calculating $I(\alpha) = \int_{\gamma(\alpha)} \operatorname{Log} \left(\frac{1-z^2}{2} \right) \cdot \frac{1}{iz} dz$, where $\gamma(\alpha)$ is the closed curve indicated by arrows in the adjoining figure, and determining $\lim_0 I$.

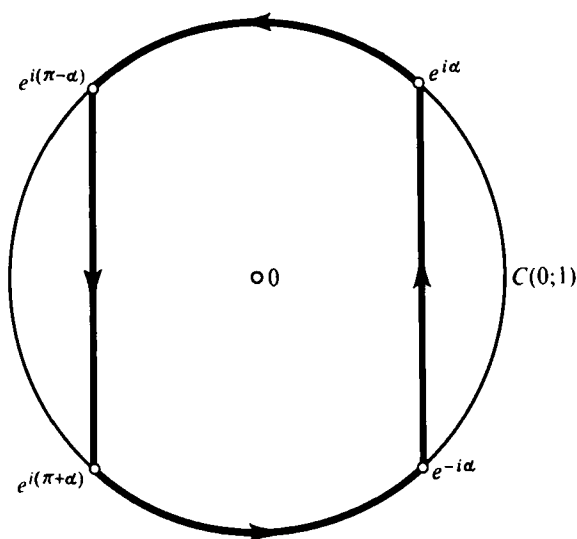


FIG. 7.2

2.7. $\int_0^\infty x^{-(\alpha+1)} \log(1+x) dx, 0 < \alpha < 1$, by using an integration by parts to show that its evaluation can be reduced to that of the integral of Example 2. [An instructive exercise is to evaluate the integral by the application of the Cauchy integral theorem and a judicious choice of path. It is possible to do this without evaluating residues.]

2.8. $\int_0^\infty t^\alpha (1-t^2)^{-1} \log t dt, 0 < \alpha < 1$. [This integral appears in the important paper on the growth of entire functions by B. Kjellberg [140].]

3. APPLICATION OF THE METHOD OF CAUCHY

We now consider representation theorems pertaining to the cotangent and cosecant which are the showpieces of the method of Cauchy. Referring to

Theorem 1.2 and using the notation β_n of Ex. 1.6, both of this chapter, we have

$$\pi \cot \pi z - \sum_{-n}^n \frac{1}{z-k} = \frac{1}{2\pi i} \int_{\beta_n} \frac{\pi \cot \pi w}{w-z} dw \quad (3.1)$$

and

$$\pi \csc \pi z - \sum_{-n}^n \frac{(-1)^k}{z-k} = \frac{1}{2\pi i} \int_{\beta_n} \frac{\pi \csc \pi w}{w-z} dw, \quad (3.2)$$

$z \in \text{int}[-(n + \frac{1}{2}), n + \frac{1}{2}; -(n + \frac{1}{2}), n + \frac{1}{2}] - \mathbf{Z}$. Since

$$\int_{\beta_n} w^{-1} \cot \pi w dw = \int_{\beta_n} w^{-1} \csc \pi w dw = 0,$$

we conclude that

$$\pi \cot \pi z - \sum_{-n}^n \frac{1}{z-k} = \frac{1}{2\pi i} \int_{\beta_n} \frac{z(\pi \cot \pi w)}{(w-z)w} dw \quad (3.3)$$

and

$$\pi \csc \pi z - \sum_{-n}^n \frac{(-1)^k}{z-k} = \frac{1}{2\pi i} \int_{\beta_n} \frac{z(\pi \csc \pi w)}{(w-z)w} dw. \quad (3.4)$$

Applying Ex. 1.6(a), this chapter, to (3.3) and (3.4) we see that given K , a compact subset of $\mathbf{C}$, there exists a positive number A such that for n sufficiently large the integrals of (3.3) and (3.4) have absolute value not exceeding A/n for $z \in K$.

We conclude that

$$\pi \cot \pi z = \frac{1}{z} + \sum_1^{\infty} \frac{2z}{z^2 - k^2}$$

and

$$\pi \csc \pi z = \frac{1}{z} + \sum_1^{\infty} \frac{2(-1)^k z}{z^2 - k^2},$$

$z \in \mathbf{C}$. It is immediate that

$$\sum_{n+1}^{\infty} \left| \frac{2z}{z^2 - k^2} \right|$$

tends uniformly to zero on a given compact subset of $\mathbf{C}$.

EXERCISES

3.1. Calculate the terms with index 1 and index 3 of the 0-Laurent sequences of $\pi \cot \pi z$ and $\pi \csc \pi z$ and determine thereby the values of

$$\sum_1^{\infty} k^{-2}, \quad \sum_1^{\infty} (-1)^k k^{-2}, \quad \sum_1^{\infty} k^{-4}, \quad \sum_1^{\infty} (-1)^k k^{-4}.$$

Represent the terms of the sequences in question of index $(2n+1)$, $n \in \mathbb{N}$, as sums of convergent infinite series.

3.2. Show that with P_n and Q_n given by

$$P_n(z) = \frac{1}{z} + \sum_1^n \frac{2z}{z^2 - k^2}$$

and

$$Q_n(z) = \frac{1}{z} + \sum_1^n \frac{2(-1)^k z}{z^2 - k^2},$$

respectively, $(P_n)_{n \in \mathbb{N}}$ tends uniformly in the sense of the chordal metric to $\pi \cot \pi z$ in $\mathbb{C}$ and $(Q_n)_{n \in \mathbb{N}}$ to $\pi \csc \pi z$ (cf. Ex. 4.35, Ch. VI).